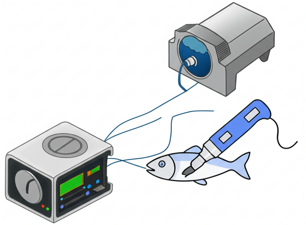
Deep Learning for Marine Biomass Analysis



Using Rapid Evaporative Ionization Mass Spectrometry

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Supervisors: Bach Nguyen
Bing Xue
Mengjie Zhang
Daniel Killeen

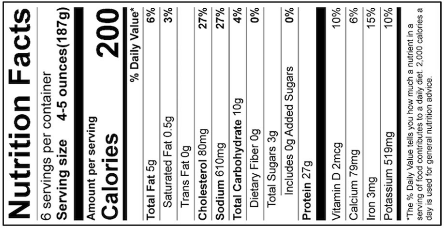
Agenda

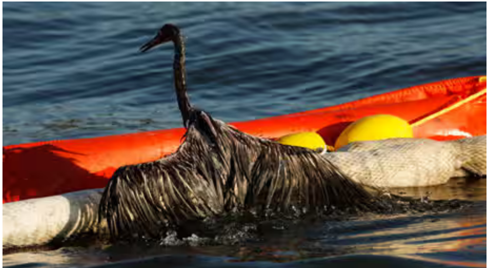
Fish $\approx$ Marine Biomass

Figure 1. Hoki (Blue Grenadier)

Figure 2. Mackerel

The Big Picture: Why This Matters?





The Core Challenge: A Lack of Trust and Safety

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Popular restaurant accused of serving cheap Vietnamese catfish to customers who thought they were getting Australian dory

- **A Melbourne restaurant has been accused of serving catfish to customers**
- **Hunky Dory has allegedly been selling frozen fillets of basa as dory**
- **Owner Greg Robotis has denied allegations he is misleading customers**
- **The City of Port Phillip is investigating Hunky Dory's Port Melbourne store**

By HARRY PEARL FOR DAILY MAIL AUSTRALIA

PUBLISHED: 14:31 AEDT, 27 May 2016 | **UPDATED:** 16:08 AEDT, 27 May 2016

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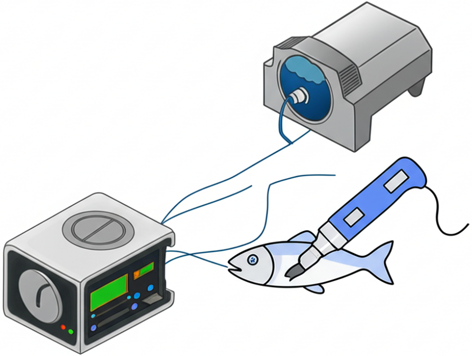
A Melbourne restaurant has been accused of serving a Vietnamese catfish to customers who believe they are ordering Dory.

A whistleblower has alleged that Hunky Dory outlets have been selling frozen fillets of basa, a species of catfish native to the Mekong basin, as fish-of-the-day dory, **The Age** reports.

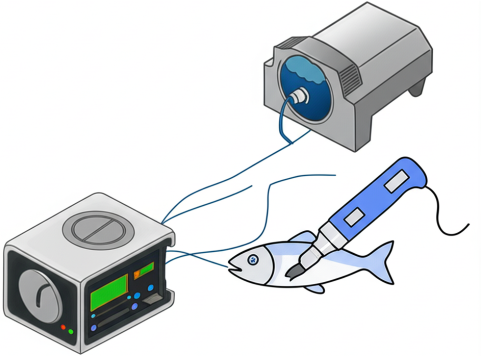
Owner Greg Robotis has denied the claims and said inexperienced staff may have been calling the fish the wrong name.

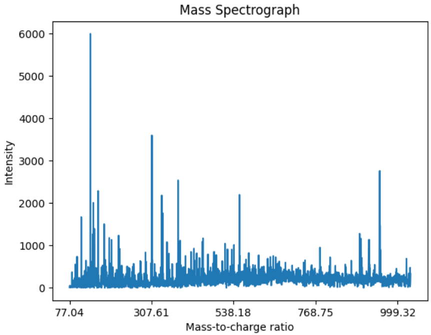
The Technical Hurdle: Identifying Samples

A "Chemical Fingerprint": The Promise of REIMS

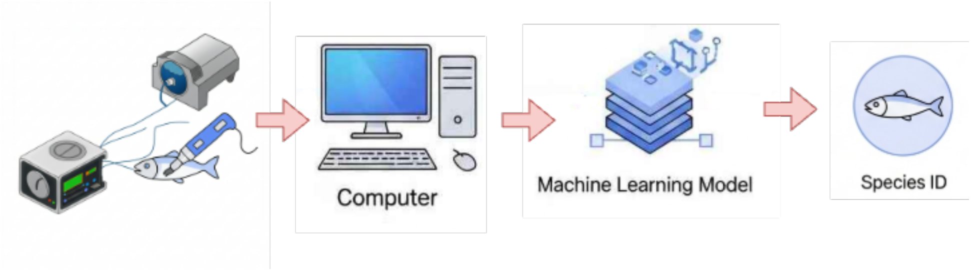


The Data Problem: Complexity & Noise

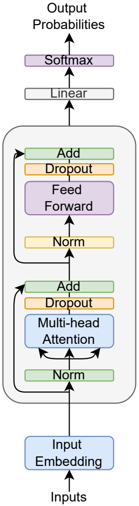




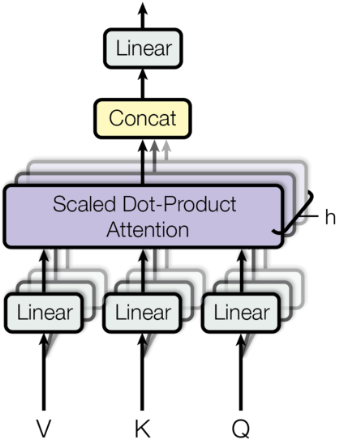
My Solution: A "Fingerprint Reader"

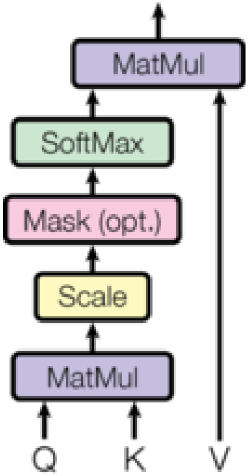


The Right Tool for the Job: The Transformer

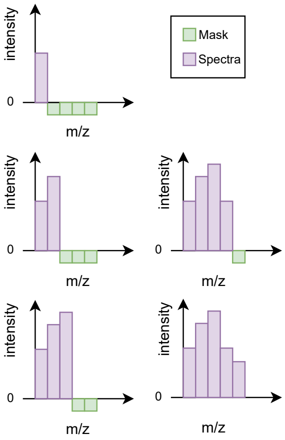


How a Transformer "Reads" Spectra

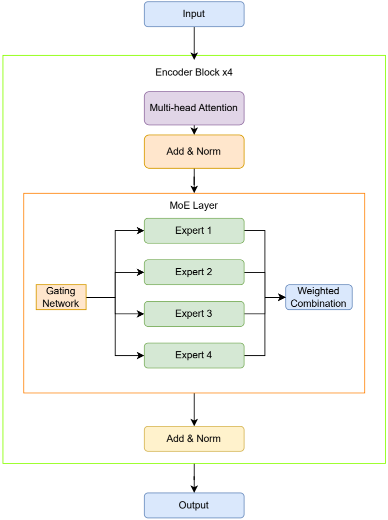




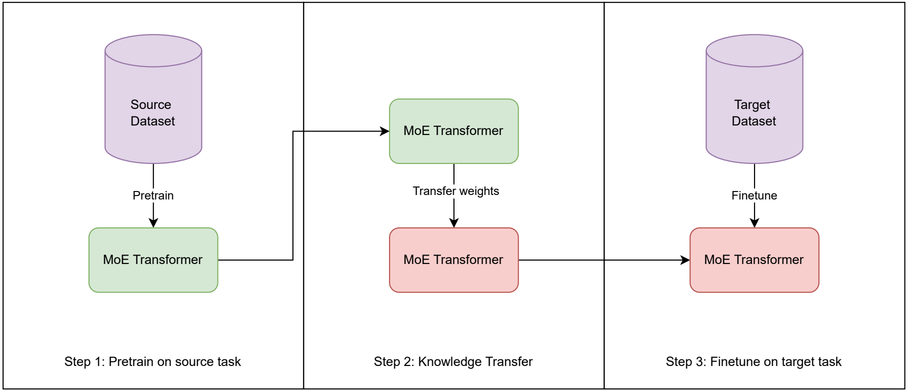
Novel Method 1: Pre-training Strategy



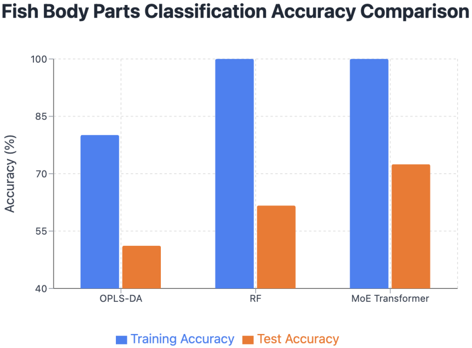
Novel Method 2: Mixture of Experts Transformer



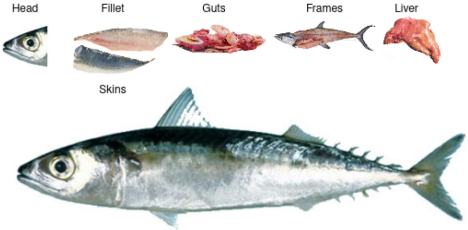
Novel Method 3: Transfer Learning for Low-Data Tasks

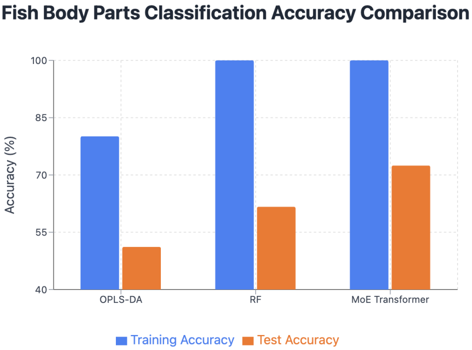


Application 1: Fish Species Identification

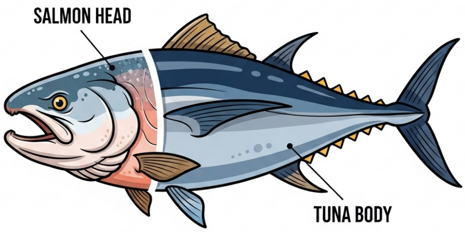


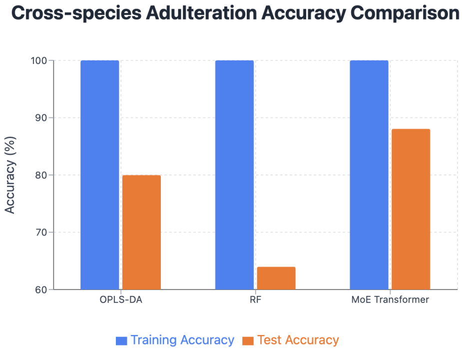
Application 2: Fish Body Part Identification



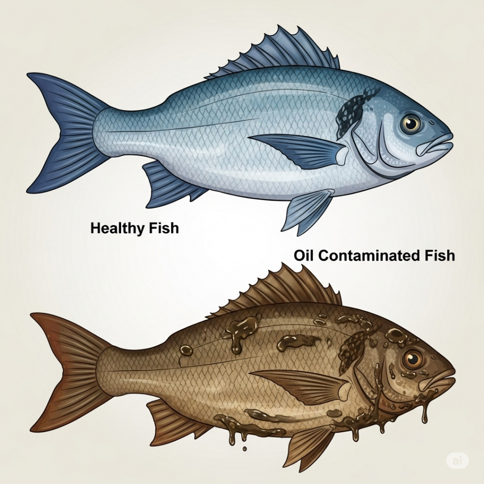


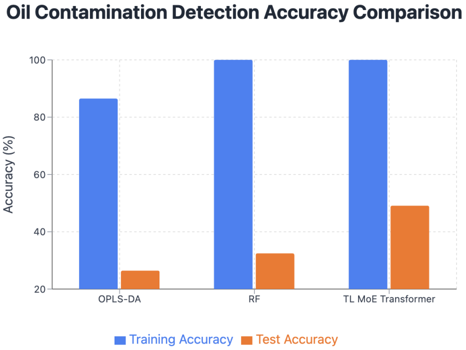
Application 3: Detecting Cross-Species Adulteration





Application 4: Detecting Oil Contamination





Interim Summary: The "Fingerprint Reader" is a Success

- ☒ Species ID (100%)
- ☒ Body Part ID (72%)
- ☒ Cross-species Adulteration (88%)
- ☒ Oil Contamination (48%)

A New Frontier: Identifying the Processing Batch

NEW ZEALAND / FOOD

Frozen anchovies in recall over biotoxin threat

6:23 pm on 25 April 2025

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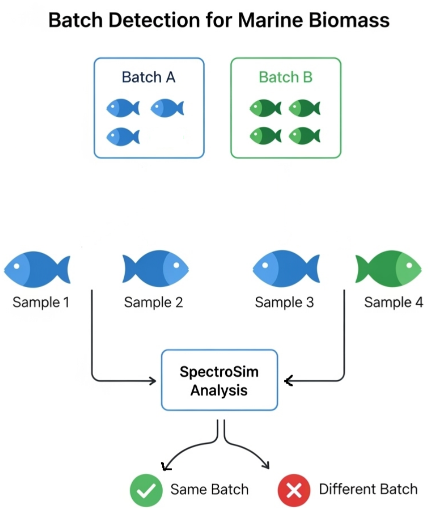
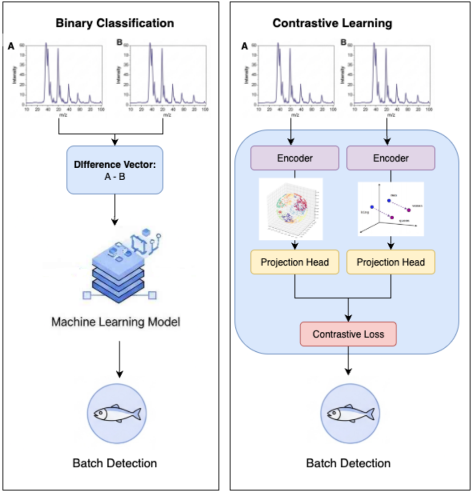
The Task: A Pair-Wise Problem



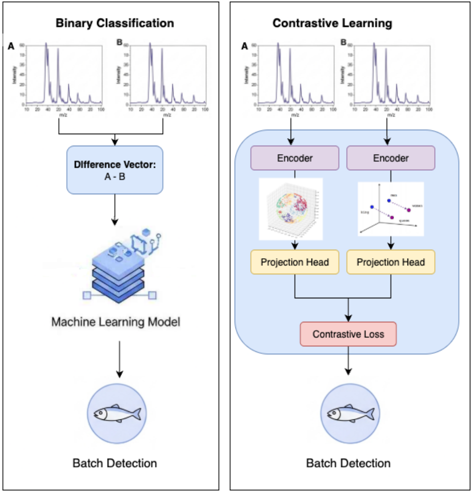
The Old Way: Binary Classification



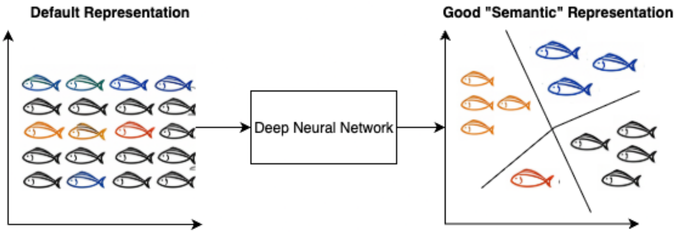
My Solution: SpectroSim

A self-supervised model to solve this without labels.

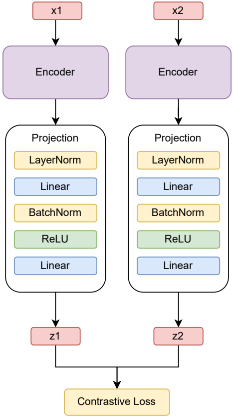
The SpectroSim Approach: Contrastive Learning

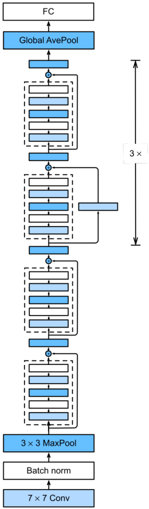


Why Contrastive Learning is Better

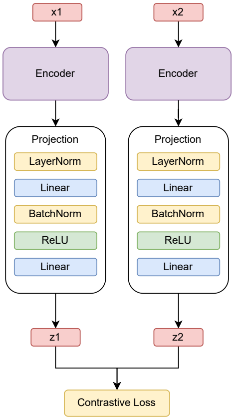


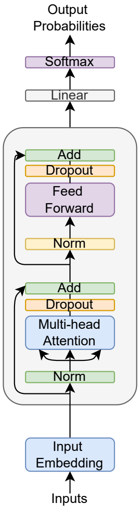
The Old Architecture of SimCLR



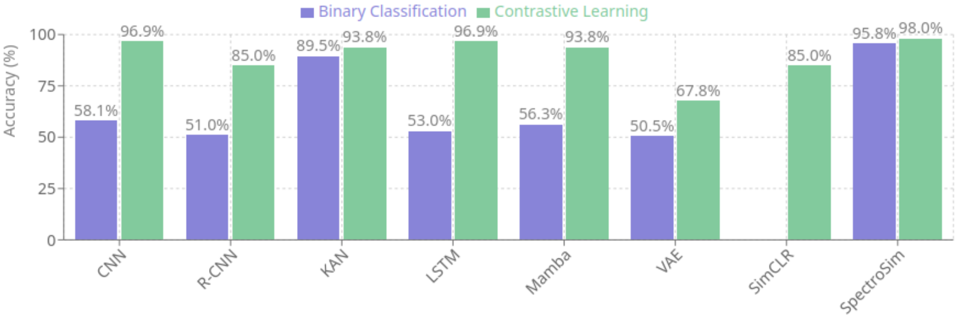


The New Architecture of SpectroSim

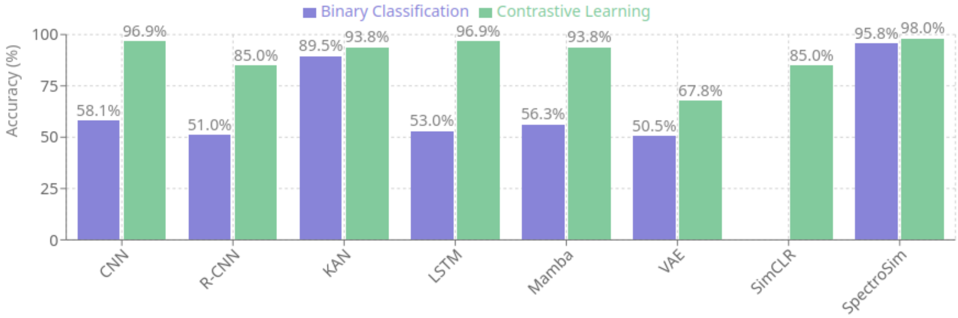




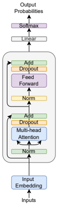
SpectroSim: The Results

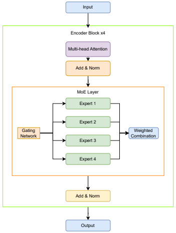


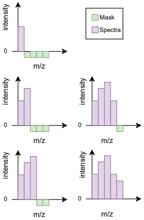
SpectroSim vs The Rest

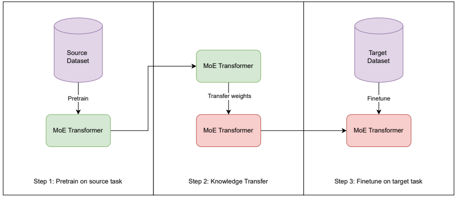


Summary of Contributions: A Better "Fingerprint Reader"

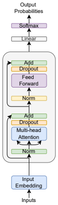


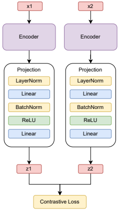


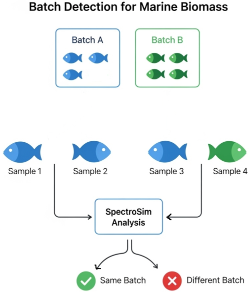




Summary of Contributions: Solving the Unsolvable



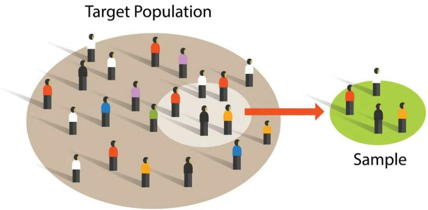


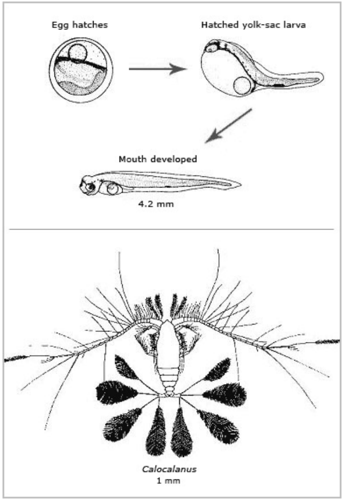


Real-World Impact

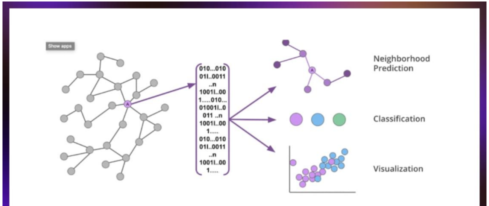
Technical Skills Demonstrated

Limitations





Future Work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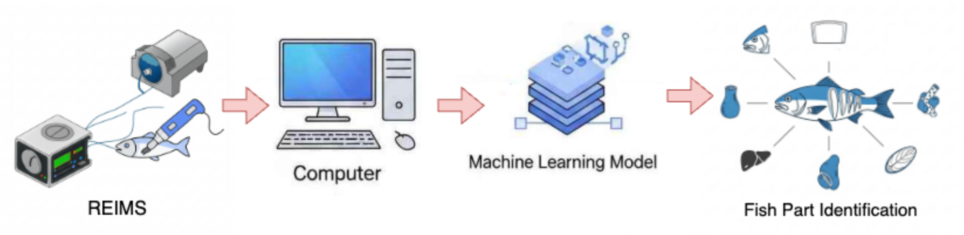
Summary

I developed a suite of novel deep learning models that can rapidly and accurately decode chemical fingerprints from fish, solving critical challenges in food safety, fraud and quality control.



Thanks for your attention!

Backup

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